

[Dashboard](#) / [My courses](#) / [MA-224-G 24H](#) / [Tests](#)/ [Test 4 \(topics 10-12: Applications of Lagrange's theorem,Coding theory,Coding theory and group codes\)](#)**Started on** Friday, 29 November 2024, 2:15 PM**State** Finished**Completed on** Friday, 29 November 2024, 2:43 PM**Time taken** 28 mins 37 secs**Marks** 2.67/3.00**Grade** 2.67 out of 3.00 (88.89%)

Information

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This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

Bob is using the RSA encryption scheme with public key equal to $(n, x) = (3139, 883)$.

a) Bob's RSA key is weak. Break his encryption and find his Bob's private key:

$y =$

Your last answer was interpreted as follows:

1387

It might help to know that the following is a list of the first prime numbers:

[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79]

b) Alice uses Bob's public key to encrypt a message, which she then sends over a public channel.

The encrypted message is

$M = 3121$

Find the unencrypted (plain text) message

$m =$

Your last answer was interpreted as follows:

3121

The following calculator can be used for computing exponents $\text{mod } (n)$:

Base:	
Exponent:	
Modulus:	
Result:	

Question **2**

Correct

Mark 1.00 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

[001111, 111111, 000101, 001000]

.

$l_{min} =$

Your last answer was interpreted as follows:

2

$l_{max} =$

Your last answer was interpreted as follows:

5

Question 3

Partially correct

Mark 0.67 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be the encoding function $E(w) = w \cdot G$, given by the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}.$$

a) Compute the minimum distance between any two codewords which are not equal.

$l_{\min} =$

Your last answer was interpreted as follows:

3

b) Find a **non-zero** codeword c for the encoding E .

$c =$

Your last answer was interpreted as follows:

10011

Any codeword will do, as long as it is not the bitstring consisting only of zeros **000...0**.

c) Find a bitstring x in $(\mathbb{Z}/2)^{\times 5}$ which is **not** a codeword for the encoding E .

$x =$

Your last answer was interpreted as follows:

10001

Syntax guide: Give your answers to **b)** and **c)** as bitstrings. E.g. if you think the answer to a question is "1011", then you should enter **1011**.

Question 4

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 412

Signing code:

Your last answer was interpreted as follows:

23053

◀ [Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)

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